

ADAI course portfolio

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Individual repository:

<https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025>

Group repository:

<https://github.com/FontysVenlo/groupepository-group-2>

Learning goal I

Start Level: Beginning

I consider myself a beginner in machine learning and neural networks, with limited theoretical knowledge and practical experience. My initial experience involved developing a basic image labeling AI model, which introduced me to concepts of artificial neural networks (ANNs) and convolutional neural networks (CNNs) in a practical context. (<https://github.com/Jazziki17/A-systems/tree/main/src>)

Weekly progress and reflections

- **Week 1:** research some information about CNNs as well as other topics discussed with the group.
(<https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/Resources/Research%20questions%20and%20answers.txt>)
Reflection: Due to a misunderstanding, I missed the lesson so I need to make sure to always check the schedule so I can attend consistently. I will make an effort to keep in contact with the group so I don't miss any information.
- **Week 2:** build a simple ai using DDI (diverse dermatology images) using TensorFlow.
(https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions/aimodel.ipynb)
Reflection: I did not have very much material prepared for the meeting and was unable to contribute much to the discussion, I was asking questions and still was part of the conversation. For the upcoming weeks I need to adequately prepare the needed tasks and make sure I am prepared.
- **Week 3:** try again to build a CNN this time starting from a pretrained model.
(https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions_V2/model.ipynb)
Reflection: As the scribe this week I was able to take some notes, however I considered my notes to not be descriptive/detailed enough so I need to focus more on the conclusion of the discussion. I will try to learn from the notes from my colleagues.
- **Week 4:** create an EDA(Exploratory Data Analysis) and do QA (quality assurance) for the presentation (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions_V2/EDA.ipynb)
Reflection: I was able to research and understand what is an EDA so I can successfully make one for the portfolio. I had to leave my QA task on the presentation till later in the

week so give me group time to add to it before I make changes, however this resulted in a lot of wasted time so I need to organize the tasks better with the group so this does not happen again.

- **Week 5:** help prepare the presentation

Reflection: I managed to discuss better task organization with the group and focus better on helping with the presentation. This week I lacked any progress on the building of the models and documenting the dataset so I need to remember to try and make more progress in this technical area. I will make task lists each week to better keep track of what I need to do

End level: Proficient

I consider to have achieved a good understanding of real-world application of machine learning, ANNs and CNNs, their requirements and limitations. I also consider to have deepened my understanding of the inner workings of neural networks and machine learning and I understand there is more to learn on this topic.

Learning goal II

Start level: Orienting

I have seen the importance of accurate labels when training machine learning models. I understand that good data labelling leads to better quality and consistency.

Weekly progress and reflections:

- **Week 10:** I set up label studio and did my set of labelling for the group assignment (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/Resources/data_labels.csv)

I also completed the individual exercise.

(https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AlModel/annotation_exercise/annotation_exercise.ipynb)

Reflection: I think I managed all my work well and completed the assignments within the needed deadlines. I think I had a lack of communication with my group as we only spoke during the class hours. I think I need to keep working as I have, but I should take an initiative to start discussions in the group so we can work well together.

- **Week 11:**

End level: Beginning

I have applied a manual annotation technique using specialized software. I understand the importance of using multiple annotators to avoid bias. I have learned how to use and the importance of Cohen's Kappa when evaluating annotations in order to spot inconsistencies.

Learning goal III

Proficiency level: Undefined

Evaluate Large Language Model concepts, focusing on their working, their applications and limitations, and their application in real-world cases based on the Data Science Methodology, taking ethical considerations into account.

Learning goal IV

Proficiency level: Orienting

Evaluate transfer learning principles and their importance for improving model performance with limited data, working on projects that use pre-trained models and datasets, leading to practical methods and positive feedback on their work.

Learning goal V

Start level: Beginning

I have participated in many projects before and have made use of skills like teamwork and investigative ability. I understand there are still areas that I lack and I strive towards improvement.

Week 1:

I missed the lesson due to a misunderstanding with the schedule but I contacted my group and tried to get up to speed with all the information I found on the course.

Week 2:

We worked in a group to discuss the previous week's research questions, I did not have everything researched myself so I tried to add to the conversation in other ways, like proposing questions or discussing vague terms.

Week 3:

I was assigned as the scribe this week so I mostly focused on writing down important notes from the meeting, however this week's discussion was pretty unorganized so it was hard to focus on both writing important notes as well as following the conversation.

(Document where I wrote the notes: https://github.com/FontysVenlo/groupepository-group-2/blob/main/ADAI_Skin_Cancer_Notes.md)

Week 4:

I was assigned QA for the presentation slides, however the day before the lesson the slides were still empty so I contacted the group to remind them to upload their information.

Week 5:

I discussed organization with the group to maybe help prevent leaving everything to the last day and organize the tasks with deadlines and minimum requirements. I also informed colleagues about changes to the presentation and some quality standards to follow.

Week 10:

I tried to organize the tasks with the group but as everyone was focused on the individual assignment, I decided to do the same and we would organize at a later time via online communication.